

28th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2024)

Unsupervised Learning Methods Reveal the Complexity of User Segregation in Political Sphere in Online Communities

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Abstract

In recent decades, political user segregation in online communities has attracted a substantial amount of research attention. However, previous discussions have been limited to a dualistic axis of democrat vs republican in the context of United States. Therefore, the current study addressed the question of whether user segregation can occur beyond the dualistic axis. To answer this question, we proposed an interpretable unsupervised method measuring the degree of political user segregation, and applied it on a large dataset of Wikipedia. We found that the segregation in Wikipedia cannot be summarized merely by the dualistic axis of democrat vs republican.

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Peer-review under responsibility of the scientific committee of the 28th International Conference on Knowledge Based and Intelligent information and Engineering Systems

Keywords: political preference; user segregation; unsupervised learning methods; independent component analysis; Wikipedia;

1. Introduction

Discussions around recent political events, including the Russia-Ukraine War that started in 2022, have underscored the growing segregation between users with different political preferences (for simplicity, referred to as *political user segregation* hereafter) in online communities. Political user segregation can be attributed to various

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reasons such as defective knowledge backgrounds, biased perceptions, divergent interests, and even language barriers [1]. On platforms such as Twitter, Facebook, Reddit, and Wikipedia, users with different political preferences are strongly inclined to interact and collaborate with those who resonate with their pre-existing beliefs and seldom exchange opinions with those with heterogeneous perspectives [2–4]. Recommendation systems assembled by online communities are likely to reinforce the degree of segregation through automated algorithmic suggestions based on similarities between users, intentionally or unintentionally exacerbating the formation of echo chambers and filter bubbles [5–7]. The acceleration of political user segregation in the cyber world is perceived as posing substantial threats to democracies, which fundamentally rely on individuals' exposure to and comprehension of a diverse array of political viewpoints [8–10]. Thus, a substantial amount of research attention has been drawn to quantitatively demonstrate the degree of user segregation using machine-learning based methods and online behaviour data [11–16].

However, most of previous discussions on political user segregation are United States-focal and have been limited to a dual axis representing the democrats and republicans in United States. To be more specific, the datasets used in previous research were based on priorly chosen subsets of discussions on ideologies and political practices of democratic and republican party in United States, and their quantitative methods depended on manually generated labels designed to address segregation within this dual axis. To this end, the questions of whether user segregation also occurs beyond the United States focused dual axis and how the degree of segregation could be measured within such axes remain open.

To investigate political user segregation beyond the democrat *vs* republican axis, we borrowed the ideas from Nature Language Processing (NLP) field. Previous studies in this field have found that interpretable semantic dimensions can emerge from word embeddings through matrix decomposition methods such as principal component analysis (PCA) and independent component analysis (ICA) [17, 18]. Following this idea, we proposed an approach to extract interpretable axes of user segregation from node embeddings of the collaboration network of users. To apply our approach, we amassed a large dataset ranging from 2001 to 2022, which records nearly 20 million edit histories on English Wikipedia related to 79,679 articles (also called entries) under the category of “Political Ideologies” and investigate the main axes in which political user segregation occurred. We find that political user segregation on Wikipedia is more complex than can be summarized by merely using the dualistic axis of democrat *vs* republican axis. Beyond this axis are other axes, such as communism *vs* anti-communism, secularism *vs* caesaropapism, cosmopolitanism *vs* nationalism, and environmentalism, which also marked significant political user segregation. In the dynamic term, the existence of axes beyond the dualism spectrum of democrat *vs* republican persists across the observation time span, and the degree of segregation within these axes is comparable to that in the democrat *vs* republican axis.

The current study expands discussions on political user segregation in online communities beyond the democrat *vs* republican axis and provides a more comprehensive description of the landscape of political user segregation on Wikipedia. In this regard, our results underscore the complexity of political user segregation and the necessity for further research on how political user segregation beyond the democrat *vs* republican spectrum affects the formation of echo chambers and filter bubbles in cyberspace. Methodologically, our proposed framework provides a balance between the independence of manually generated labels and the interpretability of results, which enables future studies to explore the extensive amount of online community data in a more efficient and insightful manner.

2. Related work

2.1. Quantifying Political User Segregation

The increasing segregation of users within online communities based on their political preferences has become a prominent topic of discussion in recent decades. Consequently, a significant amount of research has been dedicated to quantitatively demonstrating and monitoring the level of segregation among users in these communities. In this context, efforts have been made to propose unsupervised methods for detecting user segregation, utilizing approaches such as graph theory and neural networks [14, 19]. These methods fall under the broader category of community detection. While the unsupervised nature of these techniques allows for a flexible exploration of political user segregation in the digital world, they often lack interpretability. In other words, these methods do not provide

insights into the specific factors (e.g., political ideologies, attitudes toward social events, or socio-economic status) that lead to user segregation within the targeted online communities.

Alternatively, methods that employ a pre-selected axis have gained popularity due to their interpretability, facilitated by researchers' prior decisions [2-7, 11-16]. These methods project each entity (either a user or a sub-community) onto the chosen axis, enabling the determination of the user's proximity to certain poles (e.g., Democrats or Republicans) within that axis. While this approach allows for a clearer understanding of the results, it also confines the scope of analysis. Moreover, the predominance of a unique geopolitical position and the rapid polarization of domestic politics often result in a focus on the United States. The democrat vs republican dichotomy has thus been a common axis for analyzing user segregation and political polarization within various online communities, including Wikipedia [2, 20], Reddit [4, 15-16, 21], Twitter [2, 15, 16], Facebook [13], news platforms [5-6], and Amazon.com [11].

However, as hubs of the global Internet, online communities offer insights into international geopolitical events beyond the confines of U.S. domestic politics [22-24]. The impact of global issues, which cannot be easily classified into a binary of liberalism vs conservatism, significantly influences users' perceptions and behaviors. Understanding the effect of these topics requires expanding the discussion beyond the democrat vs republican axis. Moreover, even within United States domestic politics, research has shown that divisions among citizens based on political preferences exhibit a more complex structure than the simple democrat vs republican dichotomy [25, 26]. For example, citizens sharing the same partisan preferences may still show substantial differences in their opinions on public issues and political elites.

This highlights the necessity for further research on political user segregation beyond the simple binary spectrum, focusing on how it contributes to the formation of echo chambers and filter bubbles.

2.2. *Extracting Interpretable Axis from Embeddings*

Exploring the complex dynamics of user segregation in online communities necessitates the use of unsupervised methods that transcend the limitations imposed by researchers' prior assumptions. A promising approach is the application of embedding techniques, which project the interactions between users into a high-dimensional space. Within this space, the distances between entities (be they individual users or sub-communities) serve as a measure of segregation [3, 27, 28]. However, the challenge lies in the interpretability of embedding vectors, which are difficult to contextualize within the tangible frameworks of physical, social, and political realities.

To address this, significant advancements have been made in the field of Natural Language Processing (NLP). Prior research has introduced various methodologies for deriving interpretable semantic dimensions from word or document embeddings. These methods range from tailored model designs and subsequent processing of embedding vectors to the integration of supplementary metrics such as Term Frequency-Inverse Document Frequency [17, 18, 27]. Notably, one study [17] demonstrated that semantic axes, distinguishable by themes such as food or vehicles, could be identified within word embeddings through the application of matrix decomposition techniques like Independent Component Analysis (ICA). This methodology, distinct from others that rely heavily on the relationships between documents-words relationship in textual data, offers a versatile framework that can be adapted to explore embeddings in other potential domains.

Building on this insight, the current study investigates the feasibility of using ICA to uncover interpretable axes of political user segregation on Wikipedia. This approach aims to bridge the gap between the abstract representations of user interactions and their real-world implications, thereby enhancing our understanding of political divisions within online platforms.

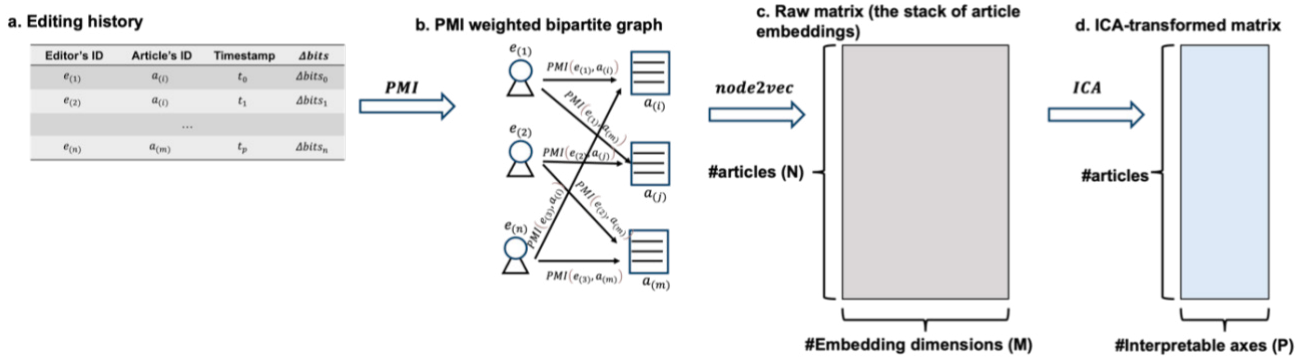


Fig. 1. The illustration of analyses (a) editing history dataset; (b) PMI weighted bipartite graph; (c) the stack of article embeddings; (d) the ICA-transformed matrix of embeddings.

3. Methods and Materials

3.1. Dataset

To apply our proposed approach, a dataset for 18,839,919 edit histories of 79,679 articles under the category of “Political ideologies” or its subcategories was collected from Wikipedia. The observation timespan is from January 21st, 2001 to December 31st, 2022. Each edit history is a timestamped record documenting how many bits of a Wikipedia article’s content were changed through one edit from a certain Wikipedia editor (see Fig. 1a). The edits from the bot and anonymous editors were excluded from the dataset.

We utilized data from Wikipedia instead of other popular online communities such as Reddit, X (formerly Twitter), Facebook, or YouTube for two reasons. First, as we show in subsequent sections, Wikipedia data provide unique labels that help us effectively validate and interpret our results. Second, unlike other communities that utilize machine-learning-based recommendation systems [13-16], Wikipedia does not use a recommendation system to guide editors to collaborate with each other based on their similarities. Thus, using the Wikipedia dataset allowed us to exclude the potential impacts of algorithms, thereby focusing on user choices as a potent force in shaping online community structures. Hence, our findings are more likely to reflect political user segregation arising naturally from online behaviour.

3.2. Measuring User Segregation Through Article Embeddings

Aligning with previous studies [4,14], we measured user segregation in our Wikipedia dataset through the editing relationships between editors and political articles. To be more specific, we first applied an expansion of node2vec [28] to learn the embeddings of Wikipedia articles from a bipartite graph (see Fig. 1b) reflecting the editing relationships of Wikipedia editors and articles, and then identified the dimensions of user segregation through the learned article embeddings.

Considering the fact that some edits may involve significant changes to an article (e.g., an essential contribution to the content), whereas others may only entail minor modifications (e.g., format revision), the edges between editors and articles in the bipartite graph were weighted by the contributions of the edits. The contribution of an edit is measured by the pairwise mutual information (PMI) based on its number of bit change. Mathematically, PMI is defined as follows [29]:

where $\overline{PMI(u, a)}$ is the PMI between an editor e and an article a . $\Delta bits(e, a)$ is the total number of bits changed by editor e on article a . E and A are the whole collection of editors and articles in the dataset. Intuitively, its value indicates whether the contributions of editor e on article a compared to a random case where editors randomly contribute the bits change to each article, given the number of bits change conducted by each editor and received by each article. If the bit change is significant between e and a , the value of PMI will be positive, otherwise that value will be negative. We then deleted the edges with negative PMIs in the bipartite.

Then, we carried out the node2vec method on this PMI-weighted bipartite graph to learn the embeddings of each Wikipedia article. Since the original node2vec model was designed for unipartite graph, we adjusted it to learn the embeddings of Wikipedia articles by a small revision. In the original node2vec model, random walk is used to enumerate an array of nodes as the ‘context’ of each target node, and then the embedding of the target node is learned through the context using a similar method in word2vec [28]. In this study, we also carried out the original random walk in the PMI-weighted bipartite of editors and articles, but then removed the nodes representing editors from the context, and only left nodes of articles. Through this way, all articles were embedded in a high dimensional space. In this space, the small distance between articles indicates that these articles shared the same group of main contributors. Conversely, the large distance between articles indicates that the editors contributed to these articles were segregated. In this regard, each subgroup with small inner-group distance but large inter-groups distance in the high-dimensional space intuitively indicates a collection of articles whose editors significantly segregated from the community. The common topic shared by this subgroup then reflects a direction which concentrated a main component of user segregation across the whole space. Thus, it can be considered as a dimension of the user segregation in Wikipedia.

It is worth noting that since the hyperparameters of the node2vec mode significantly affect its results, we conducted a grid search on the combinations of the common choices of hyperparameters in practice [30]. Eventually, the optimized hyperparameters with the number of embedding dimensions as 200, random walk length as 20, window size as 3, p as 0.5 and q as 0.5 were addressed via the topological feature construction task [30]. Additionally, to validate the robustness of the article embeddings, we repeatedly trained the node2vec model 10 rounds. We found the cross-correlations between all pairs of the embeddings in different rounds were larger than 0.83 (p -value < 0.001). We then generated robust embeddings for articles by averagely aggregating the embeddings learned in the 10 rounds.

3.3. Extracting Interpretable Dimension of User Segregation

To extract the interpretable dimension of user segregation, we then used ICA to transform the robust embeddings of Wikipedia articles. Specifically, we first stacked the embeddings of Wikipedia articles as a matrix (see Fig. 1c) with N rows and M columns, where N represents the total number of articles in the dataset (i.e., 79,679), M represents the number of embedding dimensions (i.e., 200). Then, we transformed the original matrix using fastICA algorithm as well as PCA as a preprocessing step to generate a new matrix with N rows and P columns (see Fig. 1d) [17]. Since the results of fastICA algorithm include random component, we repeatedly conducted ICA ten rounds, and matched the columns of the ICA-transformed matrix in each round through a greedy strategy based on their Pearson correlation coefficients. Finally, we averagely aggregated the matched rows to generate a robust ICA-transformed matrix.

Following research in NLP field [17], we expected that each column of this ICA-transformed matrix indicates an independent axis of user segregation, and these axes can be easily interpreted by summarizing the contents of typical articles with top-K component values in each column. We validated this assumption through two approaches. First, we projected the geometry structure of Wikipedia articles in the high dimensional space into a 2-D scatter plot using their embeddings and t-SNE algorithm [31]. Then we marked the locations of the typical articles with top-100 component values. As explained in section 3.2, the subgroups of article in the high dimensional space represents the dimension of user segregation. Therefore, if the columns of the ICA-transformed matrix represent the dimension of user segregation, a mirror relationship should be observed between the locations of the typical articles and the subgroups.

In addition to this intuitive approach, we validated the content consistency of the typical articles in each column. Since the user segregation discussed in this study rooted in the different political preference, it is arguable that each dimension of the user segregation should relate to a coherent topic in political sphere. Thus, if a column of the ICA-transformed matrix represents a dimension of user segregation, the contents of its typical articles should be consistent to some extents. We examined the consistency of the contents of representative articles in each column, through a simplified *intrusion task* [17]. In the original intrusion task, a metric measuring the content consistency will be evaluated based on the number of intruders under a large k (e.g., 100). We simplified this task as increasing the value of k until an intruder is detected. For example, given the top four representative articles in a column are about feminism, gender equality, women's right, and communism, the value of k without intruder was defined as three, since the top three articles are consistently related to gender quality and the fourth article is not. This value of k was referred to as k_{max} hereafter. The content consistency of articles was defined by their *common category tree*. The category tree is a tree-like structure reflecting the belonging relationships of articles' categories (e.g., 'Gender equality' belongs to 'Civil rights and liberties' which further belongs to 'Social liberalism') [32]. If two articles share a common category, all parent categories of this common category will also be shared by these two articles. Thus, this common category tree reflects the content consistency between articles from their general scopes to specific topics. An intruder in a column was considered as detected when the common category tree of the typical articles become void. Observing the common category tree for each axis, allowed us to easily interpret the location of the corresponding axis in the political sphere. Because the size of the table was too large to be reported in the manuscript, we reported the common category tree of all 48 detected axes in Dataset S1 as a supplementary file.

3.4. Identifying Axes Beyond the Dualism Spectrum of Democrat vs Republican

To investigate whether there were other axes beyond the dualism spectrum of democrat vs republican, we searched whether any category in the common category tree of axis contained democrat- or republican-related keywords (i.e., "democrat," "republican," "liberal," "conservat," "right," and "left"). Note that the common category tree includes all parent categories of the articles. As a result, "democrat" was included in the common category tree of articles by politicians and scholars on politics. We defined those axes whose common category tree including at least one of the democrat- or republican-related keywords as *dualism spectrum-related axes*, and calculated the percentage of these axes. To validate the distinction between the dualism spectrum-related axes and others, we further extracted the articles the titles of which included the keywords "democrat," or "republican," and compared their medians of the component values in the two types of axes using Mood's median test [33]. The component values in each axis were normalized using a maximum-minimum scaler.

We then investigated the degree of segregation within the dualism spectrum-related axes and other axes. Even if axes beyond democrat vs republican exist in Wikipedia, there is a probability that the degree of segregation within the democrat vs republican axes is much larger than that within other axes. In this respect, we used the mean of the normalized component values to measure the degree of segregation along each axis and examine whether there is statistical difference between them.

Finally, we examined the proportion of political user segregation axes across the years. Specifically, we divided our Wikipedia dataset into 21 subsets based on the year of the editing histories (from 2001 to 2022). We detected the axes of political user segregation and compared the number of the dualism spectrum-related axes with those of other detected axes, as well as the degree of segregation within different axes.

4. Results

Fig. 2 shows the visualization of the geometry structure of article embeddings in the high-dimensional space. Each grey plus character represents a Wikipedia article. When the cut-off of variance explained ratio was set as 99% for PCA as the pre-processing step, the ICA-transformed matrix then yielded 84 columns. The coloured nodes represent the typical articles with top 100 highest component values in each column. Different colours indicate the various dimensions of the typical articles. It can be found that most of the typical articles in a column were coherently located around the centre of a certain subgroups of articles. In this regard, we can intuitively consider that

the typical articles in the columns of the ICA-transformed matrix implied the dimension of the user segregation in Wikipedia.

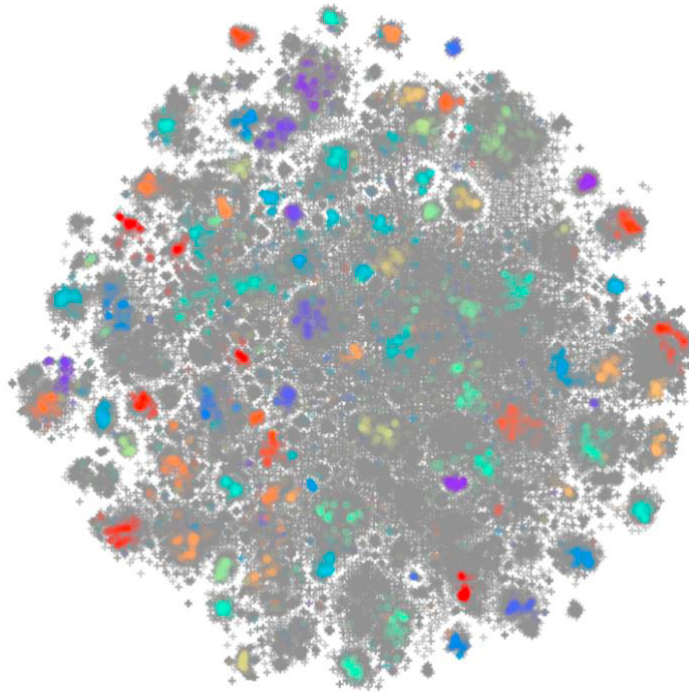


Fig. 2. Illustration of the geometry structure of Wikipedia articles and the typical articles in the high-dimensional space.

Considering the content consistency, the median of k_{max} of the 84 dimensions was 6 with the 25th percentile as 3 and 75th percentile as 17. Aligning with the previous study [17] where the content consistency of the top five representative articles (i.e., $\sqrt{k_{max}} \geq 5$) was used as a threshold to define robust dimensions, 48 dimensions were detected. This small number of axes is consistent with the expectation suggested by the previous study [17] that the high-dimensional space is a sparse one with only a limited number of prominent dimensions.

By observing the common category tree of each dimension, we can then easily interpret how the dimension is related to topics in the political sphere. We identified 15 of the 48 (31.3%) detected axes through the the democrat-or republican-related keywords. Assuming that a detected axis is a dualism spectrum-related axis, as in a Bernoulli experiment, the 95% confidence interval of the fraction of dualism spectrum-related axes is from 18.1% to 44.4%, which means that more than half of the axes of political user segregation in Wikipedia did not lie in the spectrum of democrat and republican. Then by comparing the component values of the articles whose titles included the keywords “democrat,” or “republican,” in the two types of axes using Mood’s median test, we found that the component values of these articles had a significantly higher median in the dualism spectrum-related axes than other axes (0.324 vs 0.313, $\chi^2 = 24.65$, $p\text{-value} < 0.001$). This result validates our identification of dualism spectrum-related axes.

Next, by observing the common category tree of the left 33 axes implied that beyond democrat vs republican, we found that the prominent axes of political user segregation in Wikipedia were also related to 1) communism vs anti-communism, 2) secularism vs caesaropapism, 3) cosmopolitanism vs nationalism, 4) environmentalism, and 5) other ideologies of dictatorship or authoritarianism, such as socialism, imperialism, totalitarianism, and militarism (see specific categories in Dataset S1). Additionally, we found no statistically significant difference in the average degree

of segregation between the dualism spectrum-related axes and other axes (0.326 vs 0.318, t -value = 0.88, p -value = 0.382).

Finally, as Fig. 3 shows, the above results are qualitatively consistent across time. Except for 2010 (with a fraction of 64.29%), the fractions of dualism spectrum-related axes were less than 50% across the observation period (Fig. 3a). Additionally, there was no significant difference in the degree of segregation between the dualism spectrum-related axes and other detected axes across the entire timespan (Fig. 3b).

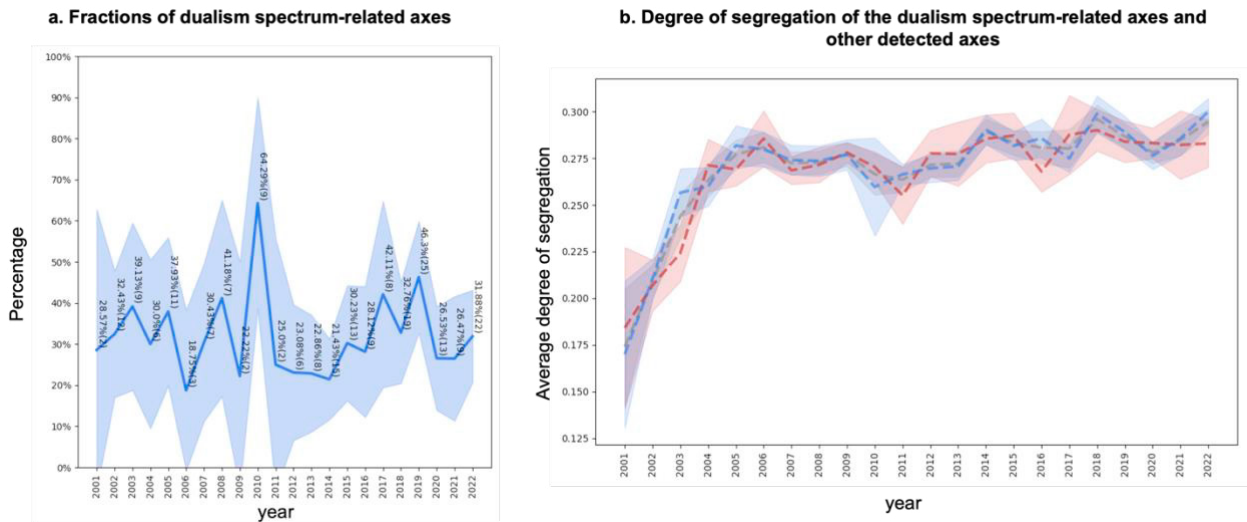


Fig. 3. (a) The percentage of the dualism-related axes in each year's sub-dataset. The line shows the estimate of the percentage, and the half-transparent area shows the confidence interval generated by two times of standard errors of the percentage. The numbers around the line indicate the specific percentages, and the numbers in parentheses show the raw number of the dualism-related axes detected by the proposed approach; (b)

The average degree of segregation across years. The lines indicate the value of the average degree, and the half-transparent area shows the confidence interval of two times of standard errors. The grey elements indicate the average degree of segregation within all axes. The red elements show the average degree of segregation within the dualism-related axes. The blue elements show the average degree of segregation beyond the dualism-related axes.

5. Discussion and Limitations

The current study addressed the question of whether political user segregation occurs beyond the democrat vs republican axis in online communities and how to measure the degree of segregation within such axes. Using Wikipedia as an example, we found that the structure of political user segregation is more complex than a dualistic spectrum of democrat vs republican. Beyond this axis are other axes, such as communism vs anti-communism, secularism vs caesaropapism, cosmopolitanism vs nationalism, and environmentalism, which also marked significant political user segregation. This result is qualitatively consistent with the dynamic term. The existence of axes beyond the dualism spectrum of democrat vs republican persists across years, and the degree of segregation within these axes is dynamically comparable to the degree in democrat vs republican axis.

Our results underscore the complexity of the political-user segregation structure in online communities. Although in context of the United States, democrat vs republican is the most representative spectrum for its political ideologies and election practices. Nevertheless, as our study showed, online communities as the hubs of the global Internet provide information on international geopolitics events beyond domestic politics in the United States [20-22]. Attitudes toward these global topics, which can hardly be simplified into a dualism between democrat and republican, also have a considerable impact on users' perceptions and behaviours. To understand the influence of these topics, it is necessary to expand the discussion beyond the democrat vs republican axis. The results of the

current study provide empirical evidence of the concerns regarding the ignorance of axes beyond democrat vs republican. In this regard, our results highlight the necessity for more research on political user segregation beyond the democrat vs republican spectrum and how it affects the formation of echo chambers and filter bubbles.

The proposed approach offers a balanced strategy for user segregation detection between the independence of manually generated labels and the interpretability of the results. Even in communities without labels similar to Wikipedia articles, our method allows researchers to easily interpret results based on the representative content detected by matrix decomposition. For instance, when applied the proposed method on the Reddit dataset [4], interpretable axes were described by the names of subreddits with top component values in each column of the ICA-transformed matrix. In this case, eight distinct axes were identified, including three beyond the democrat vs republican axis, which were linked to capitalism vs socialism, communism, and politics of specific regions (e.g., the United Kingdom, China, etc.). Previous studies proposed unsupervised methods for user segregation detection based on graph theory or neural networks [14, 19]. However, owing to their lack of interpretability, these methods have not received significant attention in subsequent research. Instead, methods with an a priori chosen axis have been widely used because researchers can interpret results with the assistance of their prior decisions [2-7, 11-16]. However, the prior chosen axis also confines the scope of analyses. In this regard, the balanced approach proposed in this paper is expected to enable future studies to explore extensive online community data in a more efficient and insightful manner. In addition to this contribution, methods in previous research [2-7, 11-16] considered all user interactions (e.g., making comments in Reddit, uploading videos on YouTube, and submitting posts on Facebook) to be equivalent. The methods proposed in this study can flexibly handle users' inequivalent interactions, allowing future studies to measure online user segregation at a higher resolution and investigate how different types of interactions affect the landscape of online information exchange.

Finally, we conclude by noting the limitations of the study. First, unlike interest-based communities, such as Reddit, Wikipedia is a knowledge-co-production community. Thus, it is arguable that the formation of the political user segregation observed in this study was the result of a lack of knowledge. Namely, if an editor is not familiar with communist ideology in the Soviet Union, they will naturally have a high probability of being absent from discussions of communism. However, this potential attribution of user segregation does not undermine the value of our discussions on user segregation beyond democrat vs republican. Knowledge production is not just an exhibition of objective facts but also involves the formation of consensus and compromise on controversial points [34-36]. This interactive aspect of the production process is particularly important for gaining knowledge in the political sphere [3]. In this respect, political user segregation rooted in knowledge sparsity can lead to topics in the political sphere that are exclusive to experts and cast even more substantial threats to democracies worldwide than those rooted in divergent interests. Therefore, knowledge-based user segregation deserves further attention in future studies. Future studies could expand the discussion to purely interest-based communities and compare their results with ours. This comparison may provide insight into the differences in user segregation rooted in different attributes. Second, the current study only analysed the vector representations of Wikipedia articles and left the vector representations of editors untouched. By analysing how segregation occurs from the perspective of editors, future research can expand their discussions to other important issues, such as how echo chambers rooted in user segregation further affect the perceptions and behaviours of online users and may be able to develop further methods to predict and even interfere with the formation of user segregation.

Data and Code Accessibility

The raw data and the Python code created for analyses in the current study, are available in a dedicated OSF repository:

https://osf.io/4er7n/?view_only=a58431698cdb446db337fba14df66a25

Acknowledgements

This study was supported by the Japan Society for the Promotion of Science KAKENHI Grant Number JP23K17010 and the HIRAKU-Global Program funded by MEXT's "Strategic Professional Development Program for Young Researchers."

References

- [1] Van Dijck, J. 2013 *The culture of connectivity: A critical history of social media*. Oxford University Press.
- [2] Flamino, J., Galeazzi, A., Feldman, S., Macy, M. W., Cross, B., Zhou, Z., Serafino, M., ≈ Bovet, A., Makse, H. A. 2023. Szymanski, B. K. Political polarization of news media and influencers on Twitter in the 2016 and 2020 US presidential elections. *Nature Human Behaviour*, 7(6), pp. 904-916.
- [3] Shi, F., Teplitskiy, M., Duede, E., Evans, J. A. 2019. The wisdom of polarized crowds. *Nature Human Behaviour*, 3(4), pp. 329-336.
- [4] Waller, I., Anderson, A. 2021. Quantifying social organization and political polarization in online platforms. *Nature*, 600(7888), pp. 264-268.
- [5] Flaxman, S., Goel, S., Rao, J. M. 2016. Filter bubbles, echo chambers, and online news consumption. *Public opinion quarterly*, 80(S1), pp. 298-320.
- [6] Hannak, A., Sapiezynski, P., Molavi Kakhki, A., Krishnamurthy, B., Lazer, D., Mislove, A., Wilson, C. 2013. Measuring personalization of web search, in: *Proceedings of the 22nd international conference on World Wide Web*, pp. 527-538.
- [7] Hou, L., Pan, X., Liu, K., Yang, Z., Liu, J., Zhou, T. 2023. Information cocoons in online navigation. *IScience*, 26(1), 105893.
- [8] Baron, D. P. 1994. Electoral competition with informed and uninformed voters. *American Political Science Review*, 88(1), pp. 33-47.
- [9] Downs, A. 1957. *An economic theory of democracy*. Harper and Row, 28.
- [10] Lassen, D. D. 2005. The effect of information on voter turnout: Evidence from a natural experiment. *American Journal of political science*, 49(1), pp. 103-118.
- [11] Shi, F., Shi, Y., Dokshin, F. A., Evans, J. A., Macy, M. W. 2017. Millions of online book co-purchases reveal partisan differences in the consumption of science. *Nature Human Behaviour*, 1(4), 0079.
- [12] Süßflow, M., Schäfer, S., Winter, S. 2019. Selective attention in the news feed: An eye-tracking study on the perception and selection of political news posts on Facebook. *New Media & Society*, 21(1), pp. 168-190.
- [13] Bakshy, E., Messing, S., Adamic, L. A. 2015. Exposure to ideologically diverse news and opinion on Facebook. *Science*, 348(6239), pp. 1130-1132.
- [14] Martin, T. 2017. community2vec: Vector representations of online communities encode semantic relationships, in: *Proceedings of the Second Workshop on NLP and Computational Social Science*, pp. 27-31.
- [15] Bail, C. A., Argyle, L. P., Brown, T. W., Bumpus, J. P., Chen, H., Hunzaker, M. F., Lee, J., Mann, M., Merhout, F., Volfovsky, A. 2018. Exposure to opposing views on social media can increase political polarization. *Proceedings of the National Academy of Sciences*, 115(37), pp. 9216-9221.
- [16] Iyengar, S., Hahn, K. S. 2009. Red media, blue media: Evidence of ideological selectivity in media use. *Journal of communication*, 59(1), pp. 19-39.
- [17] Yamagiwa, H., Oyama, M., Shimodaira, H. 2023. Discovering universal geometry in embeddings with ICA. *arXiv preprint arXiv:2305.13175*.
- [18] Park, S., Bak, J., Oh, A. 2017. Rotated word vector representations and their interpretability, in: *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pp. 401-411.
- [19] Gaudette, T., Scrivens, R., Davies, G., Frank, R. 2021. Upvoting extremism: Collective identity formation and the extreme right on Reddit. *New Media & Society*, 23(12), 3491-3508.
- [20] Lewoniewski, W., Härting, R. C., Węcel, K., Reichstein, C., Abramowicz, W. 2018. Application of SEO metrics to determine the quality of Wikipedia articles and their sources, in: *Proceeding of Information and Software Technologies: 24th International Conference*, 24, pp. 139-152.
- [21] Sohail, S. S., Khan, M. M., Alam, M. A. 2021. An Analysis of Twitter Users From The Perspective of Their Behavior, Language, Region and Development Indices--A Study of 80 Million Tweets. *arXiv preprint arXiv:2105.10245*.
- [22] Amaya, A., Bach, R., Keusch, F., Kreuter, F. 2021. New data sources in social science research: Things to know before working with Reddit data. *Social science computer review*, 39(5), pp. 943-960.
- [23] Mason, L. 2015. "I disrespectfully agree": The differential effects of partisan sorting on social and issue polarization. *American Journal of Political Science*, 59(1), pp. 128-145.
- [24] Druckman, J. N., Levendusky, M. S. 2019. What do we measure when we measure affective polarization?. *Public opinion quarterly*, 83(1), pp. 114-122.
- [25] Greenstein, S., Zhu, F. 2018. Do experts or crowd-based models produce more bias? Evidence from Encyclopedia Britannica and Wikipedia. *Mis Quarterly*, 42(3), pp. 945-960.
- [26] Neff, J. J., Laniado, D., Kappler, K. E., Volkovich, Y., Aragón, P., Kaltenbrunner, A. 2013. Jointly they edit: Examining the impact of community identification on political interaction in Wikipedia. *PloS one*, 8(4), e60584.
- [27] Abuzayed, A., Al-Khalifa, H. 2021. BERT for Arabic topic modeling: An experimental study on BERTopic technique. *Procedia computer science*, 189, pp. 191-194.
- [28] Grover, A., Leskovec, J. 2016. node2vec: Scalable feature learning for networks, In: *Proceedings of the 22nd ACM SIGKDD international conference on Knowledge discovery and data mining*, pp. 855-864.
- [29] Neffke, F. M. 2019. The value of complementary co-workers. *Science advances*, 5(12), eaax3370.
- [30] Bonner, S., Brennan, J., Kureshi, I., Theodoropoulos, G., McGough, A. S., Obara, B. 2017. Evaluating the quality of graph embeddings via topological feature reconstruction. In *2017 IEEE International Conference on Big Data*, pp. 2691-2700.
- [31] Visual Van der Maaten, L., Hinton, G. 2008. Visualizing data using t-SNE. *Journal of machine learning research*, 9(11), pp. 2579-2605
- [32] Perez, B., Feo, C., West, A. G., Lee, I. 2012. A graph-based algorithm for categorizing Wikipedia articles. *University of Pennsylvania*.
- [33] Dixon, W. J., Mood, A. M. 1946. The statistical sign test. *Journal of the American Statistical Association*, 41(236), pp. 557-566.
- [34] Kim, S., Park, S., Hale, S. A., Kim, S., Byun, J., Oh, A. H. 2016. Understanding editing behaviors in multilingual Wikipedia. *PloS one*, 11(5), e0155305.
- [35] Funtowicz, S. O., Ravetz, J. R. 1993. Science for the post-normal age. *Futures*, 25(7), pp. 739-755.
- [36] Yang, K., Tanaka, M. 2023. Crowdsourcing Knowledge Production of COVID-19 Information on Japanese Wikipedia in the Face of Uncertainty: Empirical Analysis. *Journal of medical Internet research*, 25, e45024.